

KNOWLEDGE TRANSFER GUIDE

VSD/VSDX to PDF Preview in Oracle APEX

*Repeatable implementation, environment promotion, support and
troubleshooting runbook*

APEX

User experience

OIC

Orchestration

OCI + LibreOffice

Conversion

Purpose Enable the whole team to rebuild, promote, operate and troubleshoot the Visio preview capability without depending on one individual.

Prepared for team KT | September 2026

How to Use This Guide

This document is a build-and-support runbook. It explains the exact order of work, team responsibilities, APEX components, integration dependencies, security controls, testing strategy, promotion sequence and troubleshooting method.

Terminology correction The solution converts .vsd and .vsdx files to PDF, not PSD. The generated PDF is used only for browser preview. The original Visio file remains unchanged and downloadable.

Audience

- Oracle APEX developers
- OIC and integration developers
- OCI or cloud infrastructure engineers
- Application support team
- Security and release-management teams
- Technical leads conducting future KT or environment migration

Reading path

1. Understand the target user flow and architecture.
2. Confirm ownership and prerequisites.
3. Build and test the conversion and OIC layers independently.
4. Configure APEX only after the integration endpoint is proven.
5. Promote by environment using the controlled sequence.
6. Troubleshoot one layer at a time using the decision matrix.

1. Objective and Expected User Experience

The objective is to allow users to preview Microsoft Visio files directly inside the Oracle APEX application while preserving the original source document.

```
User uploads VSD/VSDX
|
Original file stored in OCI Object Storage
|
User clicks View
|
System generates or reuses a PDF preview
|
PDF opens inside Oracle APEX
|
Original Visio file remains available for download
```

Functional requirements

- Preserve the original .vsd or .vsdx file.
- Generate PDF only for preview.
- Reuse the existing APEX document viewer where possible.
- Do not change existing upload processing or business tables unless approved.
- Use secure service-to-service authentication.
- Support SOL_DEV, DEL_DEV, UAT and future environments.
- Show a controlled fallback when conversion is unavailable.

2. Solution Architecture

```

Oracle APEX
|   Authenticated REST request
v
Oracle Integration Cloud (OIC)
|   Invoke conversion service
v
OCI-hosted conversion runtime
|   LibreOffice headless conversion
v
OCI Object Storage
|   Original VSD/VSDX + cached PDF
v
previewUrl returned through OIC to APEX
|
v
PDF displayed on Page 638

```

Component responsibilities

Component	Primary responsibility
Oracle APEX	Receives document context, calls OIC, stores previewUrl temporarily, renders the PDF, and presents fallback messages.
Oracle Integration Cloud	Provides the secured API layer, validates and maps the request, invokes the conversion service, and normalizes the response.
OCI conversion runtime	Downloads the Visio object, runs LibreOffice headlessly, validates and uploads the PDF, then returns its location.
OCI Object Storage	Stores the original Visio file and the temporary or cached PDF preview.
APEX Web Credential	Stores the OIC service-account secret outside PL/SQL and restricts its permitted endpoint URL.

3. Ownership and RACI

Team	Responsible activities
APEX development	View link, Page 638 items and process, Web Credential reference, viewer SQL, authorization, fallback UI, APEX testing.
Integration team	OIC endpoint, mappings, authentication, activation, integration logs, conversion-service invocation and error contract.
OCI platform team	Function or container runtime, LibreOffice image, IAM/resource principals, Object Storage access, network and monitoring.
Security team	Service-account approval, credential handling, URL restriction, storage access review and rotation policy.
Release/support	Environment checklist, deployment sequencing, smoke tests, operational handover and incident routing.

Key ownership rule The APEX page is not the complete solution. The feature also depends on a valid credential, active OIC flow, running conversion service, Object Storage access and environment-specific configuration.

4. Prerequisites and Intake Checklist

Object Storage

- Namespace and bucket name
- Original-file prefix or folder convention
- PDF-preview cache prefix
- Object naming rules
- Read and write permissions
- How the preview URL is secured
- Cache retention and cleanup policy

Known reference values Existing implementation used namespace capnaop2, bucket ICT_Templates, and preview prefix vsd-preview-poc/cache/. Confirm rather than assume these values for a new environment.

Conversion runtime

- LibreOffice installed in the image
- Both .vsd and .vsdx tested
- Access to download and upload Object Storage objects
- Sufficient memory, disk and execution time
- Unique temporary directories and cleanup
- Stable JSON response contract
- Maximum file-size and timeout rules

OIC

- Integration name and activation status
- Endpoint URL and authentication method
- Approved service account
- Request and response JSON schemas
- Timeout and file-size limits
- Error-response contract and log location

APEX

- Viewer page number and source report
- Required items and target table
- Project-level authorization filter
- Modal dialog configuration
- Existing Web Credential, if any
- Content security and iframe constraints

5. Phase 1 - Validate Conversion Independently

Do not begin with APEX. First prove that representative Visio files can be converted non-interactively.

```
soffice --headless --convert-to pdf --outdir /tmp input.vsd
soffice --headless --convert-to pdf --outdir /tmp input.vsdx
```

Representative test pack

- Basic flowchart
- Multi-page drawing
- Diagram with images and connectors
- Large diagram
- Filename containing spaces
- Filename containing approved special characters
- Legacy .vsd and modern .vsdx
- Corrupt or invalid file

Acceptance criteria

- PDF is created and is not empty.
- All pages and key diagram elements remain readable.
- Conversion requires no desktop interface.
- The process returns a clear success or failure result.
- Temporary content is cleaned after execution.

Gate Do not proceed to OIC or APEX integration until conversion passes independently.

6. Phase 2 - Build the OCI Conversion Service

```
Receive request
-> validate object and extension
-> check cached PDF
-> return cached URL if present
-> download VSD/VSDX
-> convert with LibreOffice
-> validate output
-> upload PDF to cache
-> return previewUrl
-> delete local temporary files
```

Recommended request contract

```
{
  "objectName": "path/to/document.vsd",
  "bucketName": "ICT_Templates",
  "namespace": "capnaop2"
}
```

Recommended success response

```
{
  "status": "SUCCESS",
  "previewUrl": "https://objectstorage.example/.../document.pdf",
  "cached": false,
  "message": "PDF preview generated successfully"
}
```

Recommended error response

```
{
  "status": "ERROR",
  "errorCode": "VSD_CONVERSION_FAILED",
  "message": "Unable to generate PDF preview"
}
```

Implementation controls

- Never overwrite the original file.
- Reject unsupported extensions.
- Sanitize object names and prevent path traversal.
- Set file-size and execution-time limits.
- Use unique temporary folders.
- Do not return internal stack traces.
- Log a correlation ID.
- Reuse an existing cached preview where appropriate.
- Validate that the output is a non-empty PDF.

7. Phase 3 - Build and Validate the OIC Integration

```
APEX request
-> OIC REST trigger
-> validate mandatory fields
-> map request
-> invoke conversion service
-> receive result
-> normalize response
-> return previewUrl to APEX
```

Handover required from Integration Team

- OIC endpoint URL
- Authentication method
- Service-account provisioning process
- Required headers
- Request schema
- Success and error response schemas
- Timeout expectation
- Maximum supported file size

Independent OIC test cases

7. Valid .vsd request
8. Valid .vsdx request
9. Invalid object path
10. Unsupported extension
11. Missing mandatory field
12. Incorrect credentials
13. Duplicate or cached request
14. Conversion timeout
15. Large-file behavior

Gate Do not debug APEX until the OIC endpoint has been proven independently with the same request contract APEX will use.

8. Phase 4 - Configure the APEX Web Credential

Property	Value or rule
Name	OCI VSD Preview
Static Identifier	OIC_VSD_PREVIEW

Property	Value or rule
Authentication	Use the approved OIC authentication type, commonly Basic Authentication in the current implementation.
Username	Approved service account. Existing implementation used SC_OICUSR.
Password	Current approved password entered manually in the target workspace.
Valid for URLs	Restrict to the approved OIC hostname or endpoint prefix.

Security rules

- Never hard-code the password in PL/SQL.
- Never put the password in substitution strings, screenshots, chat or source control.
- Re-enter the secret after migration if credentials were exported without secrets.
- Keep the static identifier exactly aligned with the page process.
- Restrict Valid for URLs as narrowly as possible.
- Document the credential owner and rotation process.

9. Phase 5 - Configure the View Link

Required report columns

```
ID
FILE_NAME
FILE_PATH
PRJ_ID -- when project-level validation applies
```

Target configuration

```
Target page: 638
P638_NEW      = &ID.
P638_FILE_NAME = &FILE_NAME.
P638_FILE_PATH = &FILE_PATH.
```

If the viewer query uses P0_PRJ_ID, ensure it is available in session state. The View link is navigation, not authorization. Page 638 must still confirm that the current user may access the selected document.

10. Phase 6 - Create the APEX Preview Item

```
Name: P638_PREVIEW_URL
Type: Hidden
Value Protected: No
```

P638_FILE_PATH must continue to represent the original Visio object. P638_PREVIEW_URL represents only the generated PDF preview. Do not overwrite the original path.

```
P638_FILE_PATH      = original Visio object
P638_PREVIEW_URL    = generated PDF preview URL
```

Session-state safeguard Set P638_PREVIEW_URL to NULL before processing a new record so an older preview URL cannot be reused accidentally.

11. Phase 7 - Create the Before Header Process

Property	Recommended value
Name	Get VSD PDF Preview
Execution point	Before Header
Condition	Run conversion only when P638_FILE_PATH ends in .vsd or .vsdx
Credential	OIC_VSD_PREVIEW
Output	Assign the validated response previewUrl to P638_PREVIEW_URL

Process responsibilities

16. Clear any previous preview URL.
17. Validate that FILE_PATH is present.
18. Check the extension.
19. Build the request JSON.
20. Call the approved OIC endpoint.
21. Capture HTTP and response details.
22. Parse the JSON response.
23. Extract and validate previewUrl.
24. Assign it to P638_PREVIEW_URL.
25. Log a controlled technical error without exposing it to the user.

Conceptual PL/SQL template

```

DECLARE
    l_response      CLOB;
    l_request_body  CLOB;
    l_preview_url   VARCHAR2(4000);
BEGIN
    :P638_PREVIEW_URL := NULL;

    IF :P638_FILE_PATH IS NULL THEN
        RETURN;
    END IF;

    IF REGEXP_LIKE(TRIM(:P638_FILE_PATH), '\.(vsd|vsdx)$', 'i') THEN
        l_request_body := JSON_OBJECT(
            'objectName' VALUE :P638_FILE_PATH RETURNING CLOB
        );

        l_response := apex_web_service.make_rest_request(
            p_url          => '<APPROVED_OIC_ENDPOINT>',
            p_http_method  => 'POST',
            p_credential_static_id => 'OIC_VSD_PREVIEW',
            p_body         => l_request_body
        );

        apex_json.parse(l_response);
        l_preview_url := apex_json.get_varchar2(p_path => 'previewUrl');

        IF l_preview_url IS NULL THEN
            raise_application_error(-20001,
                'The conversion service did not return a preview URL.');
```

```

        END IF;

        :P638_PREVIEW_URL := l_preview_url;
    END IF;
EXCEPTION
    WHEN OTHERS THEN
        :P638_PREVIEW_URL := NULL;
        apex_debug.error(
            'VSD preview failed. File: %s. Error: %s',
            :P638_FILE_PATH, SQLERRM);
END;
```

Important This template must be aligned to the real endpoint, authentication method, request fields, response path and error schema. The safest source is the exported process from the currently working application.

12. Phase 8 - Update the Page 638 Viewer

File type	Viewer behavior
DOC/DOCX	Office Online Viewer
XLS/XLSX	Office Online Viewer
PPT/PPTX	Office Online Viewer
PDF	Existing PDF viewer or PDF.js
JPG/PNG	Direct image

File type	Viewer behavior
VSD/VSDX	Generated PDF from P638_PREVIEW_URL
Other	Controlled unsupported-file message

Conceptual VSD/VSDX SQL branch

```

WHEN REGEXP_LIKE(TRIM(d.file_path), '\.(vsd|vsdx)$', 'i')
THEN
  CASE
    WHEN :P638_PREVIEW_URL IS NOT NULL THEN
      CHR(60) || 'iframe src="' ||
      APEX_ESCAPE.HTML_ATTRIBUTE(:P638_PREVIEW_URL) ||
      '" width="1000" height="600" ' ||
      'style="border:0; max-width:100%;" ' ||
      'title="VSD PDF preview"' || CHR(62) ||
      CHR(60) || '/iframe' || CHR(62)
    ELSE
      CHR(60) || 'div class="t-Alert t-Alert--warning"' || CHR(62) ||
      'The Visio preview could not be generated. ' ||
      'Please download the original file.' ||
      CHR(60) || '/div' || CHR(62)
  END
END

```

Target-environment query

Preserve the target application's database source and authorization filters. For the DEL_DEV implementation, the relevant source was CGCT_PRJ_PROCESS_FLOWS and the viewer query included the project filter below.

```

WHERE d.file_path IS NOT NULL
  AND d.id = :P638_NEW
  AND d.prj_id = :P0_PRJ_ID

```

Items to submit

```
P638_NEW,P638_FILE_PATH,P638_PREVIEW_URL,P0_PRJ_ID
```

Use the exact list required by the query and refresh mechanism. Do not copy a SOL_DEV table name into DEL_DEV merely because the preview branch originated in SOL_DEV.

13. Phase 9 - Fallback and User Experience

Condition	User-facing result
Preview generated	Display the PDF inside the viewer.
Conversion failed	Show a friendly warning and retain original-file download.
Original file missing	Show that the requested file could not be found.
Access denied	Show a generic access-denied message without disclosing record details.
Unsupported type	Show a controlled unsupported-file message.

Do not display raw JSON, stack traces, service-account information, Object Storage internals or unhandled Oracle errors to users.

14. Phase 10 - Layered Test Strategy

Layer	Verification
1. Source record	ID, FILE_NAME, FILE_PATH and PRJ_ID are populated.
2. View link	Page 638 receives the correct selected-record values.
3. Viewer query	Exactly one authorized row is returned.
4. Web Credential	Static ID, secret and URL restriction are valid.
5. OIC request	The correct Object Storage location is sent.
6. Conversion	A non-empty PDF is created in the cache.
7. Response	A valid previewUrl is returned and parsed.
8. Direct URL	The generated PDF opens directly for an authorized request.
9. Embedded viewer	The same PDF renders in APEX.
10. Original download	The original Visio file remains downloadable.

Support rule Always prove the lower layer before debugging the next layer. A blank modal is not automatically a conversion failure.

15. Temporary Debug Method

In development only, create a temporary Static Content region to expose the relevant page values.

```
<div>
  <strong>ID:</strong> &P638_NEW.<br>
  <strong>File path:</strong> &P638_FILE_PATH.<br>
  <strong>Preview URL:</strong> &P638_PREVIEW_URL.<br>
  <strong>Project ID:</strong> &P0_PRJ_ID.
</div>
```

Observed value	Likely interpretation
P638_NEW is null	The View link did not pass the record ID.
P638_FILE_PATH is null	The report query or link did not pass the object path.
P0_PRJ_ID is null	The project filter may cause the query to return no rows.
P638_PREVIEW_URL is null	The process did not run, failed, or parsed the wrong response path.
Preview URL exists but viewer is blank	Investigate iframe/PDF viewer, encoding, response headers or browser policy.

Cleanup Delete or disable the debug region after testing. Do not expose internal URLs in production UI.

16. Troubleshooting Matrix

Symptom	Checks in order
Blank modal	Confirm P638_NEW, P0_PRJ_ID, query row, and FILE_PATH before checking OIC.
Authentication failure	Credential exists; static ID matches; username/password are current; Valid for URLs permits the endpoint.
OIC error	Request schema, required headers, object path, integration activation, runtime availability and timeout.
previewUrl is null	Inspect actual response, property name, nesting, case and HTTP status.
PDF opens directly, not in APEX	Iframe/PDF.js URL, HTML escaping, encoding, X-Frame-Options, CSP, content disposition and URL expiry.
URL contains extra text	Check that project ID, labels, debug output or closing tags are not concatenated.
Works in SOL_DEV only	Credential secret, endpoint, deployment, target table, project filter, IAM, network and bucket access.

17. Environment Promotion Runbook

APEX components that can move

- Page 638 changes
- P638_PREVIEW_URL
- Get VSD PDF Preview process
- Viewer SQL and VSD/VSDX branch
- Fallback UI
- View-link configuration

Components requiring separate configuration

- Web Credential password
- OIC endpoint and activation
- Service-account access
- Function or container deployment
- IAM policies
- Object Storage permissions
- Network rules
- Cache lifecycle policy

Controlled promotion order

26. Deploy or verify the conversion service.
27. Configure Object Storage access.
28. Deploy and activate the OIC integration.
29. Test OIC independently.
30. Import the APEX changes.
31. Create or update OIC_VSD_PREVIEW.
32. Enter the environment-specific secret.
33. Configure the environment endpoint.
34. Validate the target table and project filter.
35. Run layered technical tests.

36. Run end-to-end business testing.
37. Remove debug components.
38. Obtain technical and business sign-off.

18. Configuration and Hard-coding Rules

Avoid hard-coding environment values in Page 638. Prefer APEX Application Settings, approved lookup configuration, Remote Server definitions or another governed configuration mechanism.

```
VSD_PREVIEW_OIC_URL
VSD_PREVIEW_BUCKET
VSD_PREVIEW_NAMESPACE
VSD_PREVIEW_CACHE_PREFIX
VSD_PREVIEW_MAX_SIZE_MB
VSD_PREVIEW_TIMEOUT_SECONDS
```

Secret rule Configuration values may be externalized, but passwords and tokens must remain in an approved credential store or vault.

19. Production Readiness Checklist

Functional

- ☐ .vsd preview works
- ☐ .vsdx preview works
- ☐ Original document remains downloadable
- ☐ Existing PDF, Office and image previews still work
- ☐ Unsupported files show a controlled message

Security

- ☐ No passwords in PL/SQL, substitution strings or logs
- ☐ Web Credential uses the approved service account
- ☐ Valid for URLs is restricted
- ☐ Page 638 validates project access
- ☐ Object Storage access is least-privilege
- ☐ Preview URLs do not expose unrestricted data
- ☐ Filenames and object names are sanitized

Reliability and performance

- ☐ Cached previews are reused appropriately
- ☐ Failed conversions are logged with correlation IDs
- ☐ Timeouts and large files are handled clearly
- ☐ Temporary local files and old cache objects are cleaned
- ☐ Concurrent requests and large diagrams are tested
- ☐ Monitoring exists for OIC and runtime failures

Operational ownership

- ☐ Environment configuration is documented
- ☐ Credential rotation owner is identified
- ☐ OIC support owner is identified
- ☐ Conversion-service owner is identified
- ☐ Incident routing and rollback steps are documented

20. Team KT Delivery Script

Opening

Say this Today's KT is about how the team can rebuild, support and promote the complete capability without depending on one individual. We will cover APEX, OIC, OCI conversion, Object Storage, security, testing and troubleshooting.

Key design principle

Say this We do not attempt to display the original Visio file in the browser. We preserve it for download and generate a temporary PDF representation for preview.

Key dependency

Say this The APEX page alone is not the complete solution. The capability also depends on a valid Web Credential, active OIC integration, running conversion service and Object Storage access.

Key support rule

Say this Test one layer at a time: page items, query result, credential, OIC call, conversion, returned PDF URL and finally browser rendering.

Closing

Say this Following this sequence makes the implementation repeatable. Establish integration first, configure APEX second, and verify every layer independently. The result preserves the original file and provides a secure browser-friendly preview.

21. One-Page Implementation Runbook

39. Confirm Object Storage location, access and lifecycle.
40. Validate VSD and VSDX conversion with LibreOffice.
41. Deploy the OCI-hosted conversion runtime.
42. Configure read/write access to Object Storage.
43. Create and activate the OIC integration.
44. Test OIC independently.
45. Create Web Credential OIC_VSD_PREVIEW.
46. Enter the approved service-account secret.
47. Restrict Valid for URLs.
48. Configure the View link to Page 638.
49. Pass ID, FILE_NAME and FILE_PATH.
50. Create P638_PREVIEW_URL.
51. Create Get VSD PDF Preview Before Header process.
52. Call OIC only for VSD/VSDX.
53. Parse previewUrl.

54. Store it in P638_PREVIEW_URL.
55. Update the viewer SQL.
56. Preserve target table and security filters.
57. Display generated PDF.
58. Retain original-file download.
59. Add fallback and error handling.
60. Test each layer.
61. Remove debug components.
62. Document environment configuration.
63. Obtain technical and business sign-off.

Appendix A - Quick Support Decision Tree

```

Preview does not appear
|
+-- Is Page 638 opened with P638_NEW and FILE_PATH?
|   +-- No -> fix View link / source columns
|
+-- Does viewer query return an authorized row?
|   +-- No -> fix ID / project filter / session state
|
+-- Is P638_PREVIEW_URL populated?
|   +-- No -> check process, credential, OIC and JSON path
|
+-- Does the PDF URL open directly?
|   +-- No -> check conversion, storage access and URL validity
|
+-- Does it fail only inside APEX?
|   +-- check iframe, PDF.js, encoding, headers and CSP

```

Appendix B - KT Questions the Team Should Be Able to Answer

- Why do we generate PDF instead of displaying VSD directly?
- Where is the original file stored and how is it protected?
- Which component performs the physical conversion?
- Why is OIC used?
- What does P638_PREVIEW_URL contain?
- Why can a copied APEX page fail in another environment?
- Where is the OIC secret stored?
- How is project-level authorization preserved?
- How do we prove whether a problem is conversion or rendering?
- Which components are promoted with APEX and which are configured separately?

END OF KNOWLEDGE TRANSFER GUIDE